

parameters. It should be clear that due to sampling fluctuations, we might obtain different values of $\hat{\alpha}$ and $\hat{\beta}$ from different samples. What we have to ensure is that on the average, they represent the population parameters. Thus, the entire issue now boils down to the estimation of $\hat{\alpha}$ and $\hat{\beta}$. This is what is known as the estimation of the sample regression function and we are going to discuss it in the next section.

9.6 ESTIMATION OF SAMPLE REGRESSION FUNCTION

The commonly used procedure is the least square method that we discussed in the previous unit. To recapitulate, we have a sample of observations represented by a cluster of points in the sample space and we have to pass a regression line through this cluster in such a fashion that the sum of the squares of the deviations of the observed values of Y from their estimated values from the regression line for different values of X is minimum. We are reproducing the two relevant diagrams from Unit 8 for the sake of convenience.



Fig. 9.2: Sample Scatter-Plot

Thus, we have a scatter of sample values as shown in the above diagram and we have to pass a regression line through it. We can summarise the least square procedure for obtaining such a line now. Putting mathematically the procedure amounts to Minimize

$$\sum u^2 = \sum (Y - \hat{Y})^2 = \sum (Y - \hat{\alpha} - \hat{\beta}X)^2 \text{ with respect to } \hat{\alpha} \text{ and } \hat{\beta}.$$

Following the usual minimization procedure, we obtain two normal equations given by

$$\sum Y = n\hat{\alpha} + \hat{\beta}\sum X$$

and

$$\sum XY = \hat{\alpha}\sum X + \hat{\beta}\sum X^2.$$

Solving the two normal equations simultaneously, we obtain

$$\hat{\beta} = \frac{n \sum XY - \sum X \sum Y}{n \sum X^2 - (\sum X)^2} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$$

and

$$\hat{\alpha} = \frac{\sum X^2 \sum Y - \sum X \sum XY}{n \sum X^2 - (\sum X)^2} = \bar{Y} - \hat{\beta} \bar{X}.$$

Let us now derive an important result for the estimated slope coefficient $\hat{\beta}$.

Writing lower case letters for deviations

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} = \frac{\sum x(Y - \bar{Y})}{\sum x^2} = \frac{\sum xY - \bar{Y} \sum x}{\sum x^2}$$

But $\sum (X - \bar{X}) = 0$

$$\text{Thus } \hat{\beta} = \frac{\sum xY}{\sum x^2}$$

Now putting $k = \frac{x}{\sum x^2}$, as x 's are given.

We have

$$\hat{\beta} = \sum kY$$

Thus, $\hat{\beta}$ is a linear function of the observed values of Y . This is an important result, which we shall use later in the issue of hypothesis testing.

The following diagram shows a regression line, with the values of $\hat{\alpha}$ and $\hat{\beta}$ given by the above expressions fitted into our sample scatter.

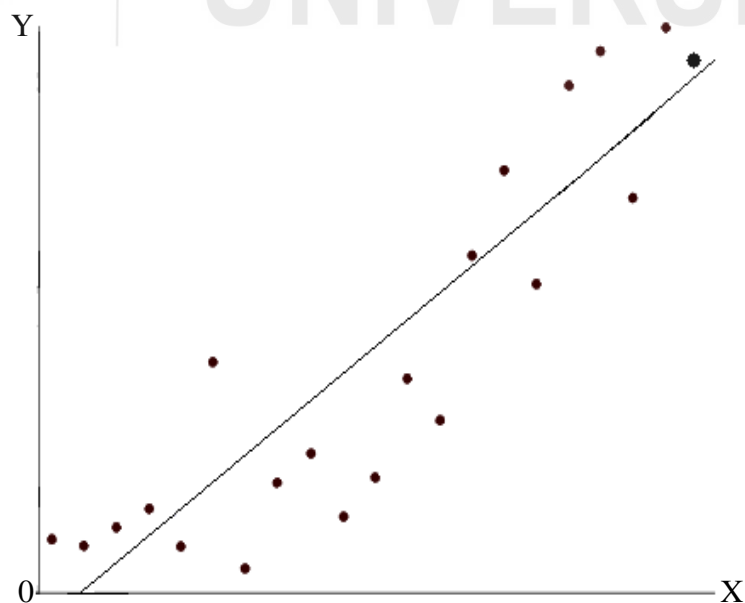


Fig. 9.3: Sample Scatter-Plot with the Sample Regression Line

It may be noted here that the sample regression line has the equation, $Y = \hat{\alpha} + \hat{\beta}X$ with the values of $\hat{\alpha}$ and $\hat{\beta}$ obtained from the above-mentioned least square method. It can be clearly seen that the values of $\hat{\alpha}$ and $\hat{\beta}$ can be calculated in terms of sample observations only and no other information that is not known is required for this purpose. As we can see, both $\hat{\alpha}$ and $\hat{\beta}$ are linear functions of X and Y . Consequently, they are called linear estimators.

Gauss-Markov Theorem

The least square estimators $\hat{\alpha}$ and $\hat{\beta}$ can be taken as the estimators of the unknown population parameters α and β because they satisfy certain desirable properties. We can state them without proofs below.

Under the assumptions of the classical regression model as mentioned earlier,

- 1) Least square estimators are linear. As we have already seen from the expressions of $\hat{\alpha}$ and $\hat{\beta}$, they are linear functions of the variables.
- 2) Least square estimators are unbiased i.e., $E(\hat{\alpha}) = \alpha$ and $E(\hat{\beta}) = \beta$. This means, if we consider different values of the least square estimates obtained from a number of random samples for a given population on the average, they will be equal to the unknown population parameters. They will not be systematically overestimating or underestimating the population parameters.
- 3) Among all the linear unbiased estimators, least square estimators have the minimum variance. In this sense, they are termed as the efficient estimators.

All these properties of the least square estimators lead to what is known as the Gauss-Markov Theorem. The theorem states:

Under the assumptions of the classical linear regression model, among all the linear unbiased estimators, the least square estimators have the minimum variance. In other words, the least square estimators are the best linear unbiased estimators or in short, BLUE.

Standard Error of the Regression Estimate

We have already stated that due to sampling fluctuations, we should expect to obtain different values of the least square estimates $\hat{\alpha}$ and $\hat{\beta}$ from sample to sample. Consequently, if we measure the standard deviations of these values of the two least square estimates from their respective expectation or mean, we shall get an idea about the extent to which these estimates are affected by the sampling fluctuations. In other words, the standard deviations of the least square estimates can be taken as a measure of the precision of these estimates. **The standard deviations of the least square estimates are known as the standard errors of the estimates.** It should be clear that the standard errors are the standard deviations of the sampling distributions of the least square estimates. The standard deviations or standard errors are obtained by taking the positive square root of the variances of $\hat{\alpha}$ and $\hat{\beta}$. The expressions for both the variance and standard of the least square estimators are given below.

$$\text{var}(\hat{\alpha}) = \frac{\sum X^2}{n \sum (X - \bar{X})^2} \sigma^2$$

$$\text{se}(\hat{\alpha}) = \sqrt{\frac{\sum X^2}{n \sum (X - \bar{X})^2} \sigma^2}$$

$$\text{var}(\hat{\beta}) = \frac{\sigma^2}{\sum (X - \bar{X})^2}$$

$$\text{se}(\hat{\beta}) = \frac{\sigma}{\sqrt{\sum (X - \bar{X})^2}}$$

It should be recollected that σ is the standard deviation of the error term U in our population regression model and we assume that it is a constant. However, since the population regression model is unknown, it follows that σ is also unknown. As a result, for the calculation of variance and standard error of the least square estimates $\hat{\alpha}$ and $\hat{\beta}$, we need to estimate σ . It can be proved that

an unbiased estimator of σ is $\sqrt{\frac{\sum \hat{U}^2}{n-2}}$. Here, $n-2$ is what is known as the degrees of freedom in statistics. This concept you must have studied in your compulsory course in statistics. Now, $\hat{U} = Y - \hat{Y}$ is the sample regression error term and accordingly we can calculate it. Thus, by replacing σ by its unbiased estimator we can compute the standard errors of both $\hat{\alpha}$ and $\hat{\beta}$. It should be noted here that we can write the unbiased estimator of σ as

$$\hat{\sigma} = \sqrt{\frac{\sum \hat{U}^2}{n-2}} = \sqrt{\frac{\sum (\hat{U} - \bar{\hat{U}})^2}{n-2}}$$

This is, because, $\bar{\hat{U}}$ is the mean of the sample regression errors. Now, while calculating it, the positive errors will tend to cancel the negative errors and as a result, it will be reduced to zero. From above expression of the estimator of σ , it can easily be interpreted as the standard deviation of the sample observations

about the estimated sample regression line. The sample estimator $\hat{\sigma} = \sqrt{\frac{\sum \hat{U}^2}{n-2}}$ is known as the standard error of estimate or the standard error of the regression.

9.7 GOODNESS OF FIT

In the earlier section, we have discussed the procedure for fitting the sample regression line and its purpose. Once such a regression line is fitted, we may be interested in examining how good has been the fit in the sense how faithfully it can describe the unknown population regression line. This is known as the issue of goodness of fit. In this matter, the regression error term or residual $\hat{U}$

plays an important role. Small quantities of residuals imply that a large proportion of variation in the dependent variable has been explained by the regression equation and consequently, the fit is good. Similarly, large quantities of residuals obviously point to a poor fit. At this stage, what we are interested in is to obtain a quantitative measure of the goodness of fit that is free of any unit for the purpose of comparability. In the previous unit, we have referred to the coefficient of determination, which is the square of the correlation coefficient. It can be shown that this coefficient of determination acts as a measure of goodness of fit. We may consider it now. The variation in the dependent variable Y about its mean can be conceptualized as

$$\text{var}(Y) = \sum (Y - \bar{Y})^2$$

We can decompose it into two components. The first is the variation explained by the regression. The second is the portion that remains unexplained by the regression. Thus we can write

$$(Y - \bar{Y}) = (Y - \hat{Y}) + (\hat{Y} - \bar{Y})$$

Squaring and applying summation on both the sides

$$\sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum (Y - \hat{Y})^2 + 2 \sum (\hat{Y} - \bar{Y})(Y - \hat{Y})$$

Now let us try to find the value of $\sum (\hat{Y} - \bar{Y})(Y - \hat{Y})$. We have the regression equation

$$Y = \hat{\alpha} + \hat{\beta}X + \hat{U}$$

$$\text{or } \bar{Y} = \hat{\alpha} + \hat{\beta}\bar{X} + \bar{\hat{U}}$$

Subtracting the second equation from the first equation

$$Y - \bar{Y} = \hat{\beta}(X - \bar{X}) + \hat{U}, \text{ because } \bar{\hat{U}} = \frac{\sum \hat{U}}{n} = 0, \text{ as } \sum \hat{U} = 0.$$

Writing lower case letters for the deviations of the variables from their mean, we have the sample regression equation in the form

$$y = \hat{\beta}x + \hat{u}$$

Now

$$\begin{aligned} & \sum (\hat{Y} - \bar{Y})(Y - \hat{Y}) \\ &= \sum (\hat{Y} - \bar{Y}) \hat{U} \\ &= \sum \hat{Y} \hat{U} - \bar{Y} \sum \hat{U} \\ &= \sum \hat{Y} \hat{U}, \text{ because } \sum \hat{U} = 0 \\ &= \sum (\hat{\alpha} + \hat{\beta}X) \hat{U} \\ &= \hat{\alpha} \sum \hat{U} + \hat{\beta} \sum X \hat{U} \quad \because \sum \hat{U} = 0 \\ &= \hat{\beta} \sum X \hat{U} \end{aligned}$$

We have

$$x = X - \bar{X}$$

$$\Rightarrow \sum x = \sum (X - \bar{X})$$

$$\text{or } \sum x\hat{U} = \sum (X - \bar{X})\hat{U} = \sum X\hat{U} - \bar{X}\sum \hat{U} = \sum X\hat{U}$$

Now

$$\sum x\hat{U} = \sum x(y - \hat{\beta}x) = \sum xy - \hat{\beta}\sum x^2$$

$$\text{We know, } \hat{\beta} = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2} = \frac{\sum xy}{\sum x^2}$$

Putting the value of $\hat{\beta}$ in the above expression for $\sum x\hat{U}$,

$$\sum x\hat{U} = \sum xy - \frac{\sum xy}{\sum x^2} \sum x^2 = \sum xy - \sum xy = 0$$

Hence,

$$\sum (\hat{Y} - \bar{Y})(Y - \hat{Y}) = 0$$

Thus,

$$\sum (Y - \bar{Y})^2 = \sum (\hat{Y} - \bar{Y})^2 + \sum (Y - \hat{Y})^2$$

If $\sum (Y - \bar{Y})^2$ is defined as the total sum of squares (TSS), $\sum (\hat{Y} - \bar{Y})^2$ as the explained sum of squares (ESS) and $\sum (Y - \hat{Y})^2$ as the residual sum of squares (RSS),

We have

$$\text{TSS} = \text{ESS} + \text{RSS}$$

Dividing both the sides by TSS,

$$\frac{\text{TSS}}{\text{TSS}} = \frac{\text{ESS}}{\text{TSS}} + \frac{\text{RSS}}{\text{TSS}}$$

Defining the ratio of ESS to TSS as R^2 , we have

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{ESS}}{\text{TSS}}$$

It should be clear now that R^2 or coefficient determination can be interpreted as the proportion of total variation in Y explained by the regression of Y on X .

Let us now consider a numerical example to fix the concepts and ideas that we have discussed so far.

Example 9.1

The hypothetical figures for the total labour force and the number of employed out of that for the period 1991-2000 are given below in Table 9.1. Run a regression of the number of employed on the total labour force.

Table 9.1: Actual Employment and Labour Force

Year	Employed (million)	Labour Force (million)
1991	100	120
1992	125	140
1993	140	165
1994	160	185
1995	175	200
1996	195	210
1997	230	250
1998	245	255
1999	270	305
2000	295	320

Let Y be the dependent variable employed and X be the independent variable labour force. The detailed calculation for working out the regression is shown below:

Year	Y	X	XY	X^2	$y = Y - \bar{Y}$	$x = X - \bar{X}$	x^2	y^2	xy
1991	100	120	12000	14400	-93.5	-95	9025	8742.25	8882.5
1992	125	140	17500	19600	-68.5	-75	5625	4692.25	5137.5
1993	140	165	23100	27225	-53.5	-50	2500	2862.25	2675
1994	160	185	29600	34225	-33.5	-30	900	1122.25	1005
1995	175	200	35000	40000	-18.5	-15	225	342.25	277.5
1996	195	210	40950	44100	1.5	-5	25	2.25	-7.5
1997	230	250	57500	62500	36.5	35	1225	1332.25	1277.5
1998	245	255	62475	65025	51.5	40	1600	2652.25	2060
1999	270	305	82350	93025	76.5	90	8100	5852.25	6885
2000	295	320	94400	102400	101.5	105	11025	10302.25	10657.5
Sum	1935	2150	454875	502500			40250	37902.5	38850
Mean	193.5	215							

$$\hat{\beta} = \frac{\sum xy}{\sum x^2} = \frac{38850}{40250} = 0.965217 \text{ and } \hat{\alpha} = \bar{Y} - \hat{\beta}\bar{X} = 193.5 - (0.965217 \times 215) = -14.217$$

Year	$\hat{Y} = \hat{\alpha} - \hat{\beta}X$	$\hat{U} = Y - \hat{Y}$	$\hat{U}^2$	$\hat{Y} - \bar{Y}$	$(\hat{Y} - \bar{Y})^2$
1991	101.8043	-1.80434	3.255643	-91.6957	8408.094
1992	121.1087	3.89132	15.14237	-72.3913	5240.503
1993	145.2391	-5.23911	27.44822	-48.2609	2329.114
1994	164.5434	-4.54344	20.64289	-28.9566	838.4821
1995	179.0217	-4.0217	16.17407	-14.4783	209.6212
1996	188.6739	6.32613	40.01992	-4.82613	23.29153
1997	227.2826	2.71745	7.384535	33.78255	1141.261
1998	232.1086	12.89137	166.1873	38.60864	1490.627
1999	280.3695	-10.3695	107.5262	86.86949	7546.307
2000	294.8477	0.15226	0.023183	101.3477	10271.36
Sum		0.00045	403.8043		37498.67

In the next stage we compute the following:

$$V(\hat{\beta}) = \frac{\sigma^2}{\sum x^2}, V(\hat{\alpha}) = \frac{\sum X^2}{n \sum x^2} \sigma^2 \text{ and } R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (Y - \bar{Y})^2}.$$

But the population variance σ^2 is unknown. So we have to use an unbiased sample estimator $\hat{\sigma}^2$ for the same. Such an unbiased estimator is given by

$$\hat{\sigma}^2 = \frac{\sum \hat{U}^2}{n-2}, \text{ where } n-2 \text{ is the degrees of freedom. Here, } n-2 = 10-2 = 8.$$

$$\text{Therefore, } \hat{\sigma}^2 = \frac{\sum \hat{U}^2}{n-2} = \frac{403.8043}{8} = 50.475537.$$

$$\text{Thus, } V(\hat{\beta}) = \frac{\hat{\sigma}^2}{\sum x^2} = \frac{50.475537}{40250} = 0.001254$$

$$s.e.(\hat{\beta}) = \sqrt{V(\hat{\beta})} = \sqrt{0.001254} = 0.0354118$$

$$V(\hat{\alpha}) = \frac{\sum X^2}{n \sum x^2} \hat{\sigma}^2 = \frac{502500}{10 \times 40250} \times 50.475537 = 63.016042$$

$$s.e.(\hat{\alpha}) = \sqrt{V(\hat{\alpha})} = \sqrt{63.016042} = 7.9382644$$

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{37498.67}{37902.5} = 0.989345$$

Finally, we present a summary of all the important results concerning the above-mentioned question.

$$\hat{\beta} = 0.965217 \quad V(\hat{\beta}) = 0.001254 \quad \text{and} \quad s.e.(\hat{\beta}) = 0.0354118$$

$$\hat{\alpha} = -14.0217 \quad V(\hat{\alpha}) = 63.016042 \quad \text{and} \quad s.e.(\hat{\alpha}) = 7.9382644$$

$$\hat{\sigma}^2 = 50.475537$$

$$R^2 = 0.989345 \quad \text{Degrees of Freedom (d.f.)} = 8$$

The estimated regression line is given by

$$\hat{Y} = -14.0217 + 0.965217X$$

It may be mentioned here that it is this kind of a summary of the results that is generally used for the further analysis of the regression model. Let us now interpret some of the results from the above summary. The regression line estimates the average employment (Y) for a given level of labour force (X). The slope coefficient $\hat{\beta} = 0.965217$ estimates the rate of change of employment with respect to labour force. For example, if 100 more persons start looking for jobs, about 97 of them actually get employed. The intercept $\hat{\alpha} = -14.0217$ can be interpreted as the average combined effect of all those variables that might also affect employment but have been omitted for the purpose of the above-mentioned regression. The coefficient of regression $R^2 = 0.989345$ indicates that about 99 per cent of the variation in employment can be explained by a variation in the labour force, which is indeed high and indicates a good fit to the given sample.

Check Your Progress 1

- 1) Discuss the meaning of linearity in the regression model.

.....

.....

.....

.....

.....

- 2) How is the regression model non-deterministic in nature?

.....

.....

.....

.....

.....

.....

3) Discuss the assumptions of the classical regression model.

.....

.....

.....

.....

.....

4) State the Gauss-Markov Theorem.

.....

.....

.....

.....

.....

5) Explain the concept of goodness of fit.

.....

.....

.....

.....

.....

9.8 FUNCTIONAL FORMS OF REGRESSION MODEL

We have already mentioned that our focus is on a linear regression model. The issue of linearity has also been considered in Section 9.2. It is in fact linear in parameter regression model that is relevant for us. However, linear in parameter regression models may also have different functional forms. We shall now briefly discuss four such regression models.

1) Linear Model

This functional form is the most common one and we have already discussed it. Its equation is given by

$$Y = \alpha + \beta X + U$$

As we know, it is both linear in parameter and linear in variable. This model can be estimated by the ordinary least square (OLS) method. The least square